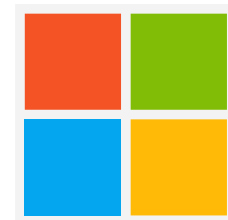


Build the unified knowledge base for agents with Foundry IQ



Phi Huynh
Microsoft MVP



Note: The slide is inherited from Microsoft's Foundry IQ team

Your People. Your Agents. Your IQ.





Foundry IQ

The knowledge layer for agents

aka.ms/FoundryIQ

Use chat history for context
(the user said earlier it was a
security fix)

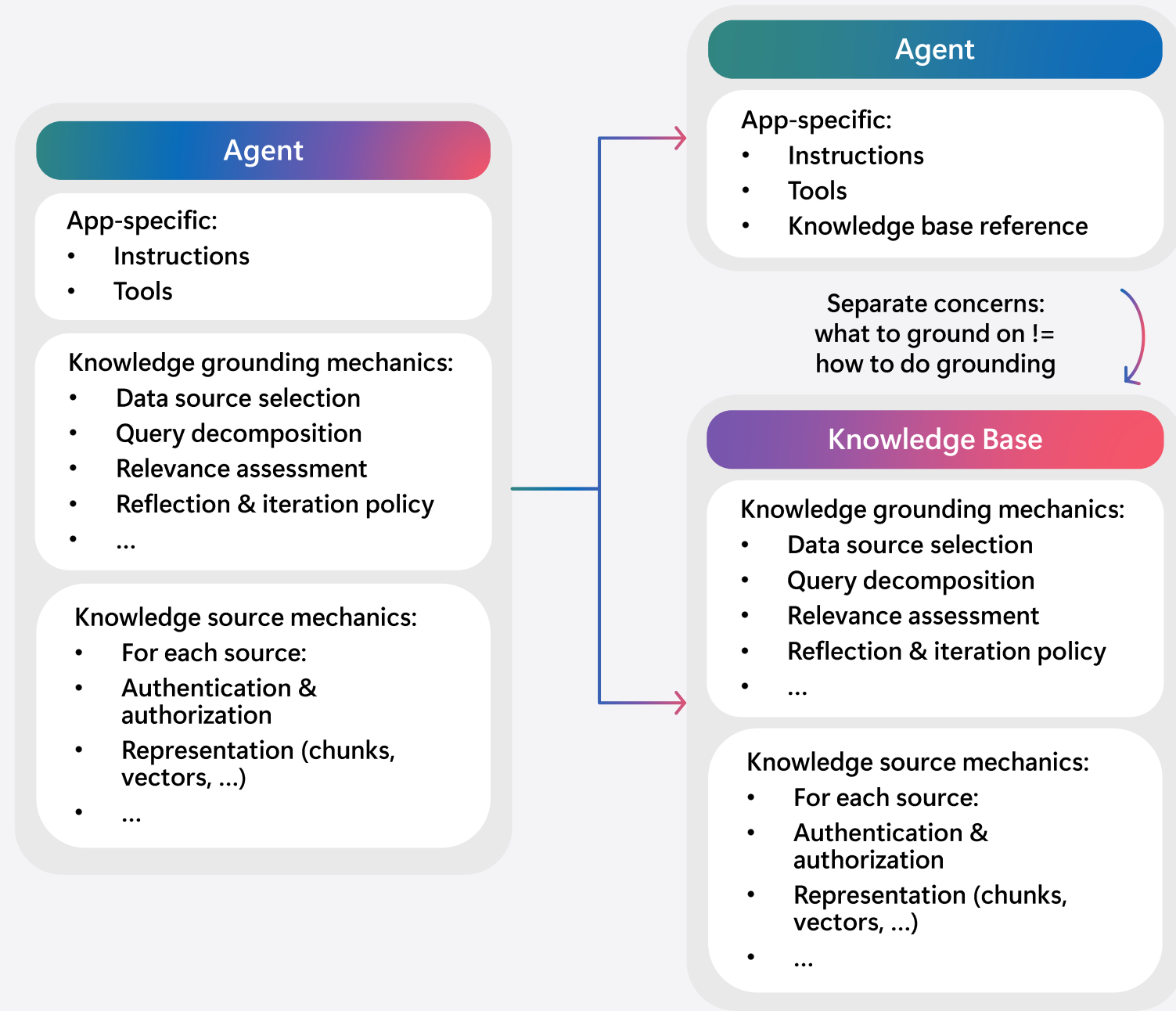
Split queries for different
information requirements,
resolving co-references

**what does KB4048959 fix and
what systems is the patch
compatible with?**

Correcting spelling is much
more effective if done in
sentence context

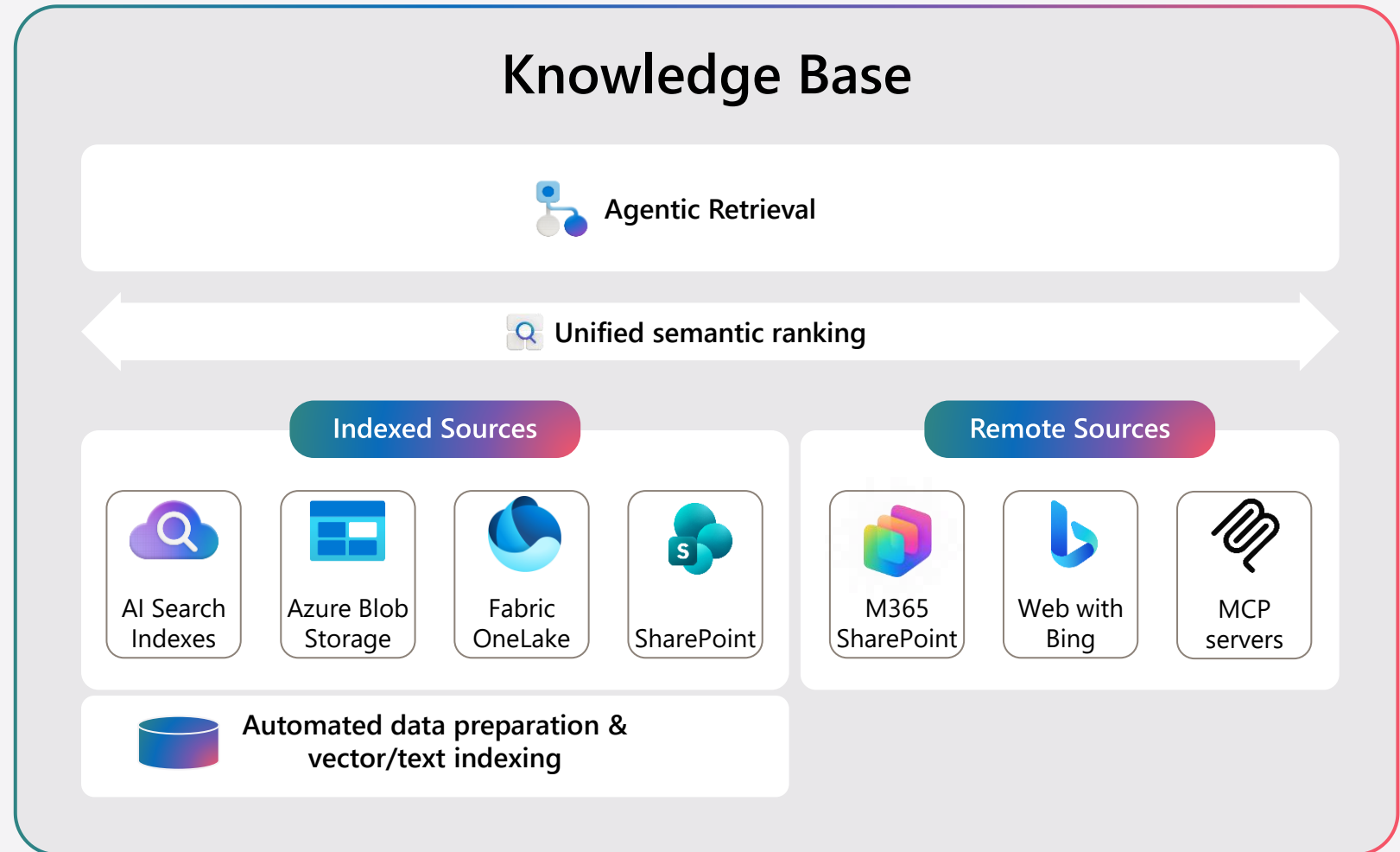
Knowledge Bases

- Represent a complete domain of knowledge
- Separate concerns between agents and knowledge retrieval



Knowledge Sources

- Encapsulate a source of data
 - Connection
 - Authentication
 - Authorization
 - Instructions
- Indexed or Remote
- All get semantic reranking boost



Foundry IQ → Azure AI Search

Foundry IQ is built on AI Search

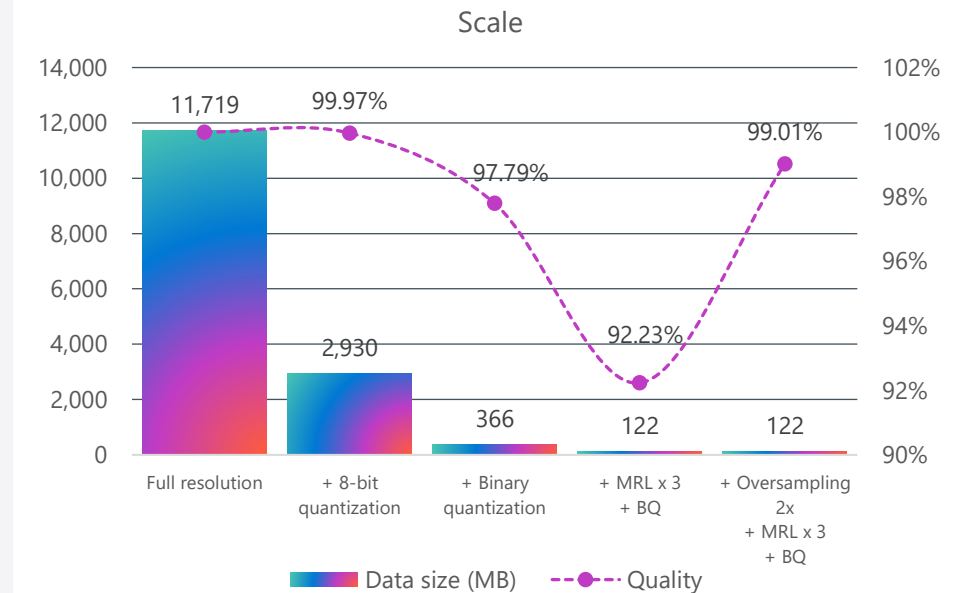
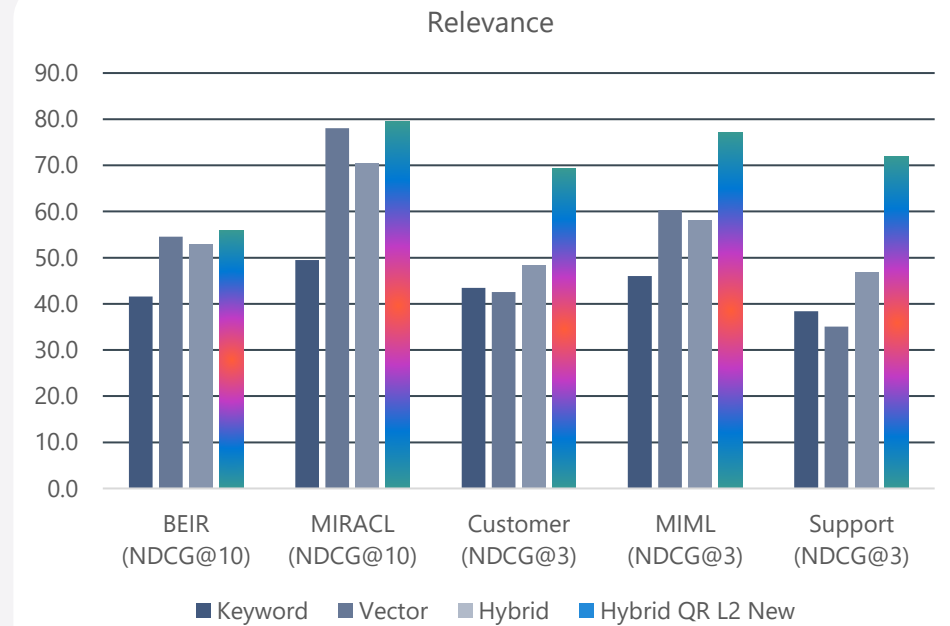
- SOTA search quality & scale

Knowledge bases usable directly in AI Search

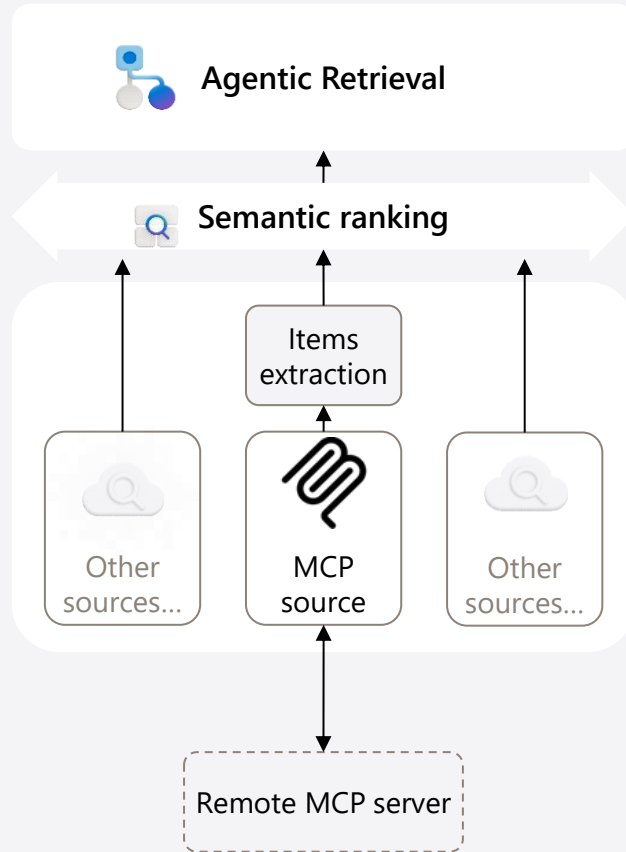
- Azure Portal & API

Works on existing AI Search services

- Add knowledge bases, connect sources
- Existing search indexes → knowledge sources



Extensibility: MCP Servers as Knowledge Sources



MCP servers: broadly available, easy to make

- Bring 3rd party sources
 - Wrap your own apps
-

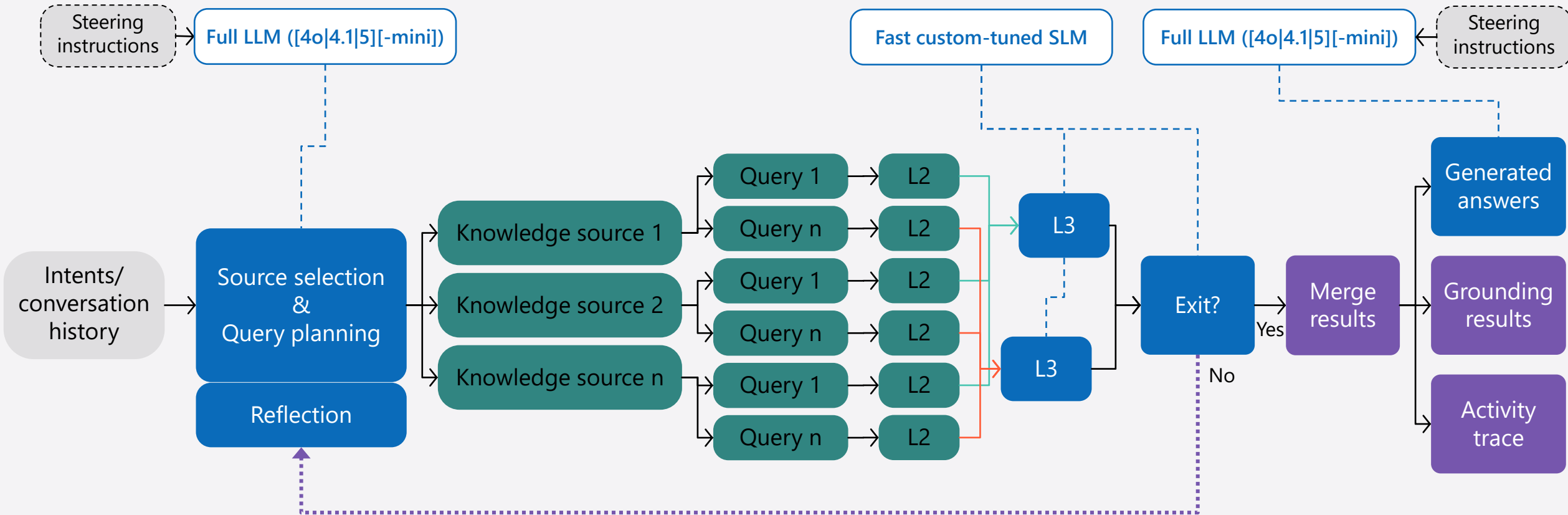
Retrieval stack boost

- Heuristically turn responses into lists of items
 - L2 reranking for improved quality
-

Private preview, sign up today

- [Aka.ms/KR-MCPsource](https://aka.ms/KR-MCPsource)

Agentic Retrieval Pipeline



Reasoning effort level →	Minimal	Low	Medium
Query planning & source selection	No	Yes	Yes
Web grounding	No	Yes*	Yes*
L2 ranking	Yes	Yes	Yes
L3 classifier	No	No	Yes
Reflection/iteration	No	No	Yes
Answer generation	No	Optional	Optional

* Requires answer gen.

Data Preparation

→Applies to indexed knowledge sources

Abstracted RAG ingestion pipeline

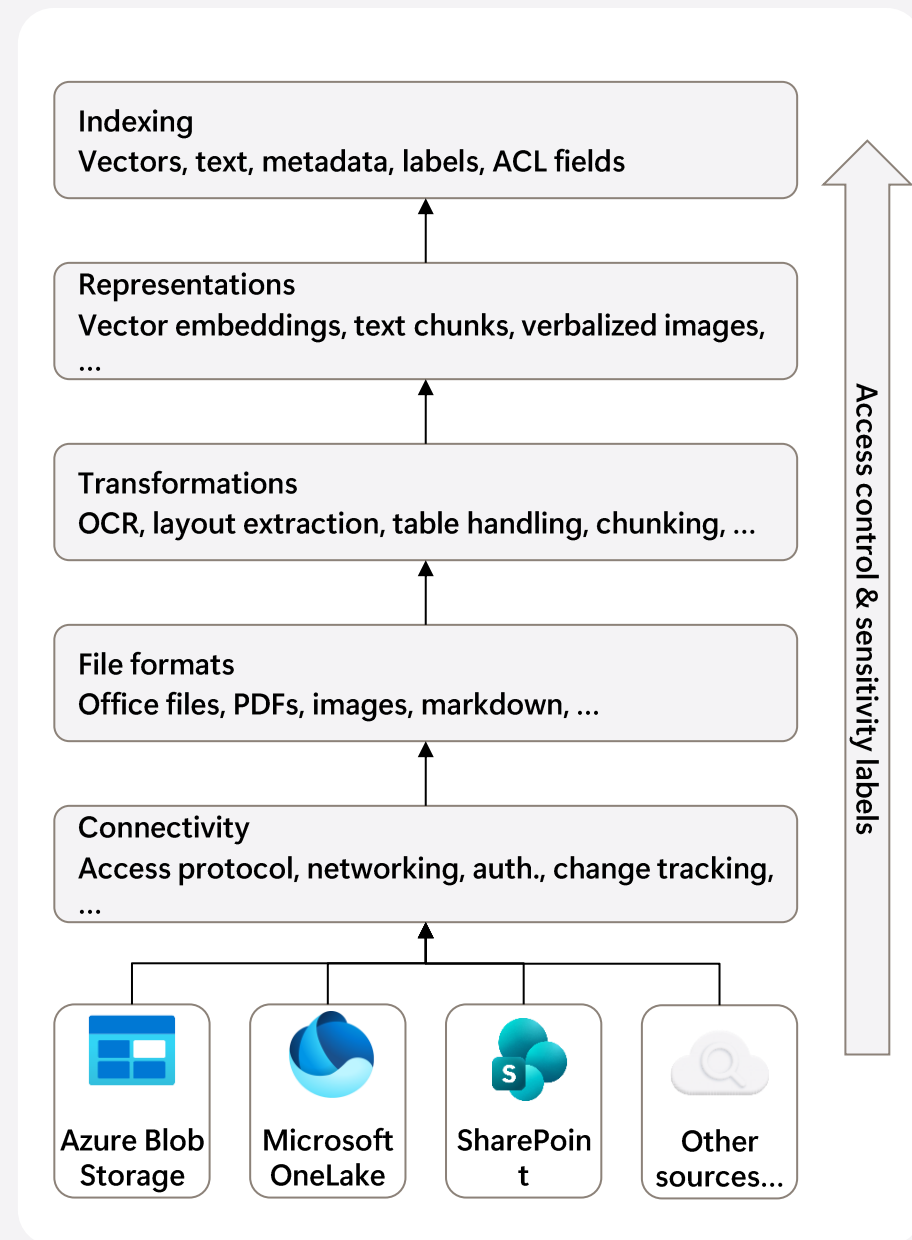
- Built on AI Search indexer stack, full control when needed

Azure Content Understanding integration

- Exceptional quality in OCR, PDF parsing, layout, tables

Multi-media support

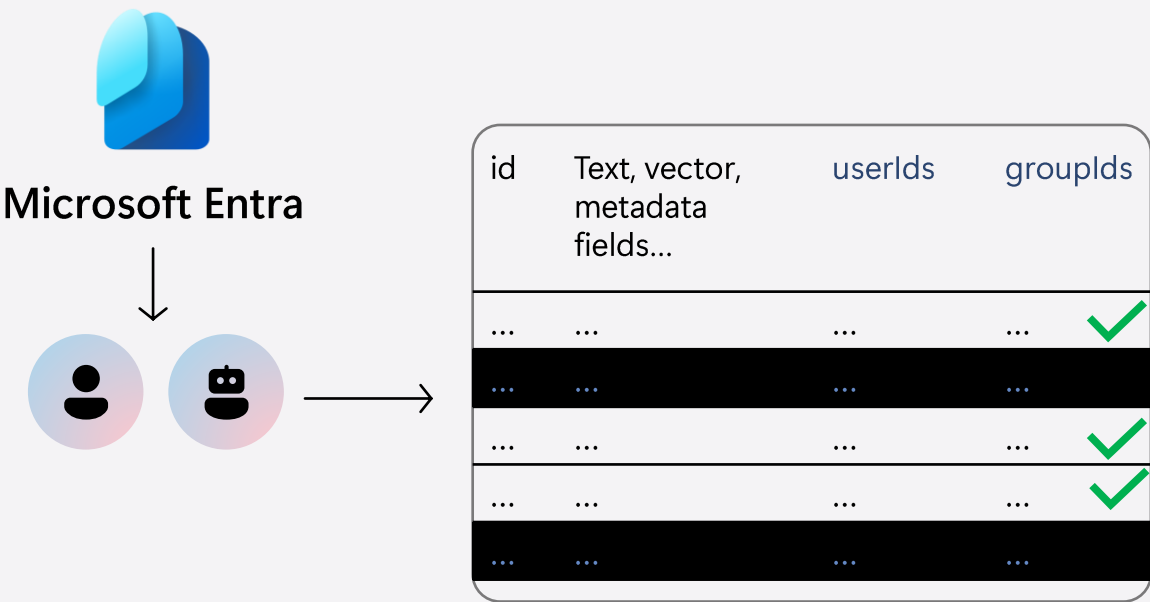
- Image extraction, embedding, verbalization



Enterprise-Grade Security

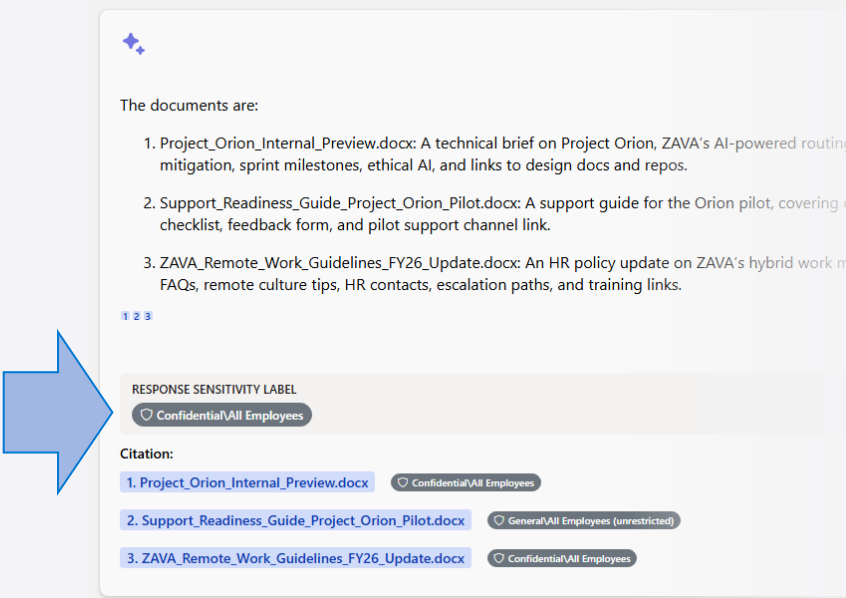
Access control

- Entra-based user identity propagation
- Document-level access control in indexes
 - SharePoint (indexed), ADLS gen2, custom indexes
- Identity propagation on remote sources
 - Connect to SharePoint (remote), uses user identity



Governance

- Integrated with Microsoft Purview
- Support for documents with sensitivity labels
- Index documents with labels, enforce policy, propagate label information

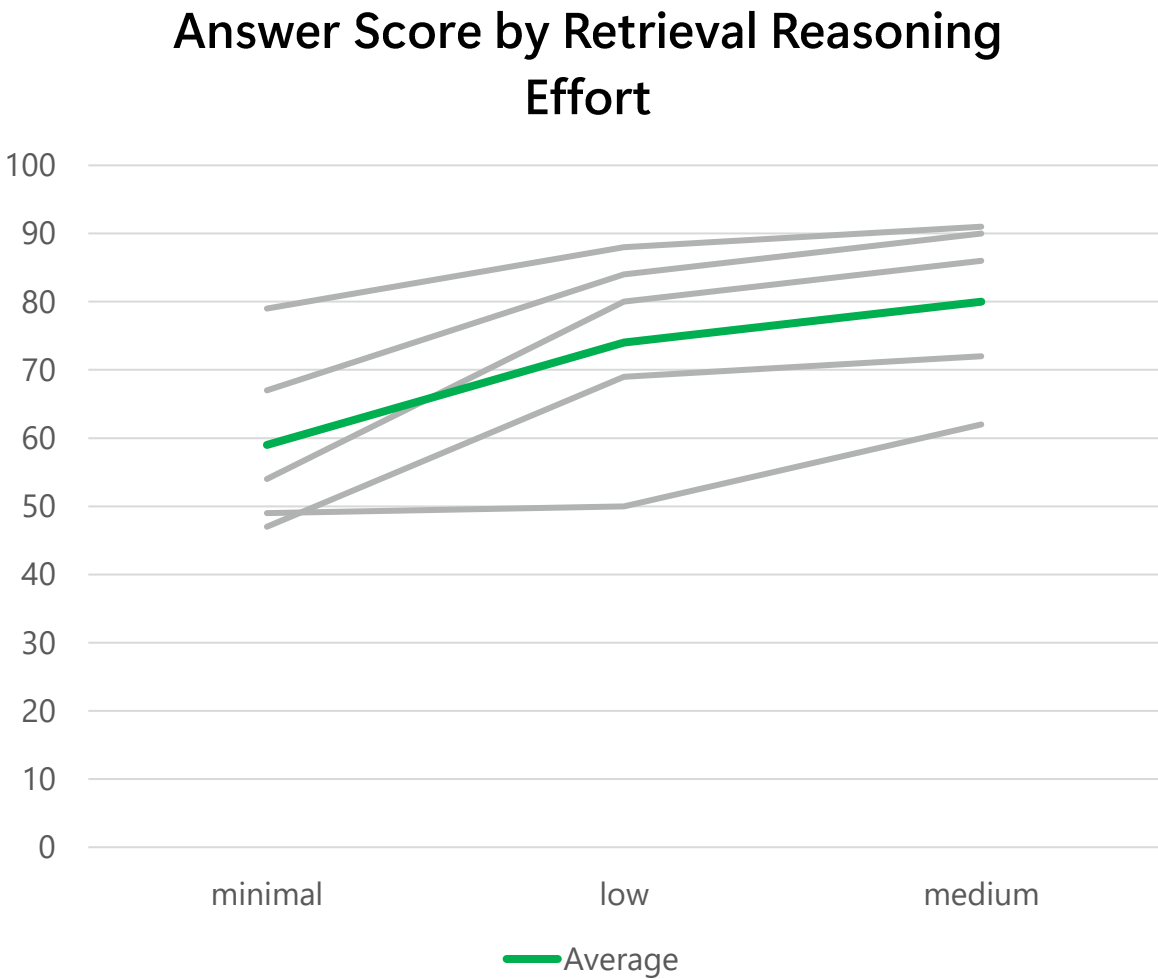


→ aka.ms/Ignite25/aisherech-purview-sensitivity-labels-repo

Evals: Is agentic retrieval actually better?

+36% Iterative search gives a 36% gain over directly searching all KSs

KB Set up		Answer Score			Minimal -> Medium	
Dataset	KS Type	minimal	low	medium	absolute	relative
sec-hardest	AZS	49	50	62	13	26%
miml-hardest-en	AZS	47	69	72	26	55%
sec-open	Web	67	84	90	22	33%
hotpot	Web	79	88	91	13	16%
frames	Web	54	80	86	32	60%
Average		55	74	80	25	36%



Get started

Resources

Learn

Blog: Introducing Foundry IQ by Azure AI Search

aka.ms/FoundryIQ

Blog: Evals

aka.ms/kb-evals

Docs: What's new

aka.ms/AISearch-new

Repos and samples

AI Search with Purview sensitivity labels

aka.ms/Ignite25/aisearch-purview-sensitivity-labels-repo

Quickstart: agentic retrieval & Foundry Agent service

aka.ms/AISearch-ar-agent

Sample app: Agentic retrieval

aka.ms/ragchat

Thank you